

Appendix A - Abrolhos Marine Park Western Rock Lobster Natural Values Report

North-west Marine Parks Network

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Natural Values

Understanding of natural values

Sampling summary

Table 1: Summary of western rock lobster pot sampling on Yamatji Shallow Bank, showing the number of pots deployed, the number of pots that fished successfully and were counted, and the number of lobster measured in each year and zone status. No-Take = National Park Zone, Fished = Special Purpose Zone.

Year	Status	Pots deployed	Pots counted	Lobster measured
2025	Fished	93	93	483
2025	No-Take	55	55	273
2026	Fished	84	84	248
2026	No-Take	55	55	124

Spatial distribution

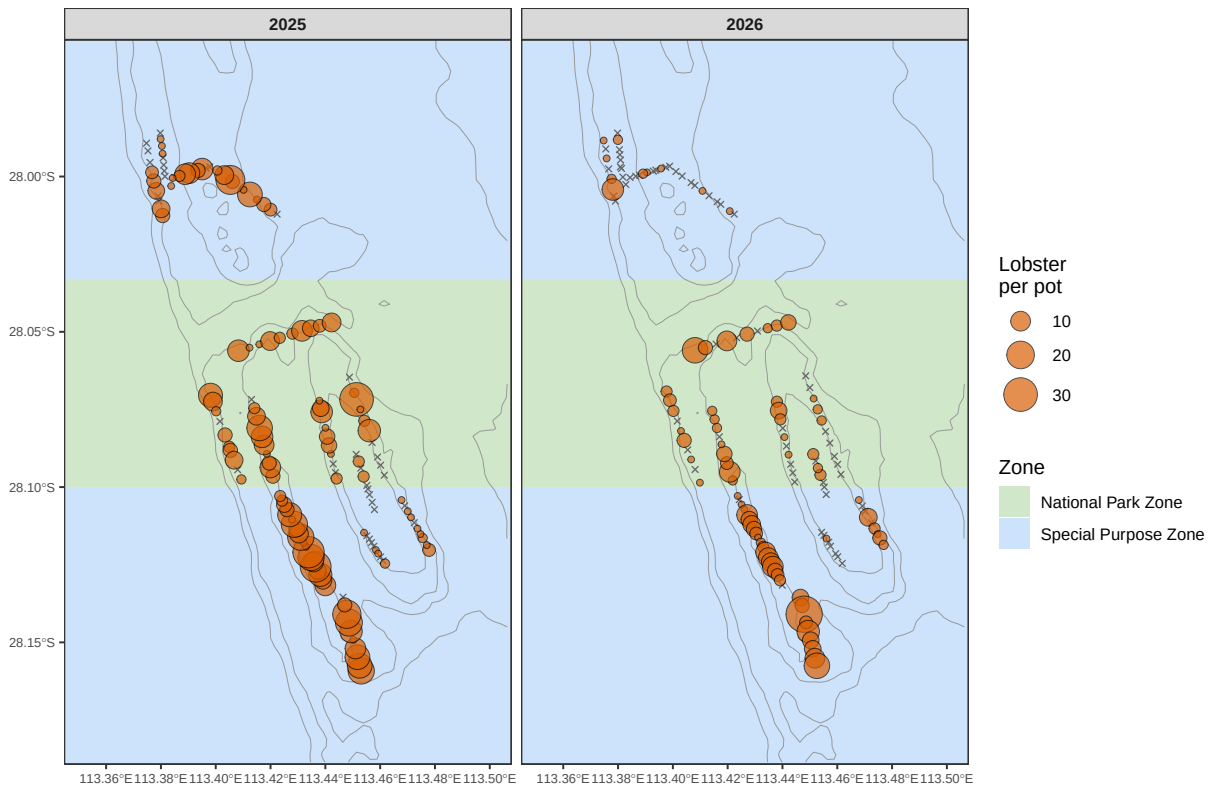


Figure 1: Spatial distribution of western rock lobster abundance per pot, shown separately for each survey year. Symbol size is scaled to the number of lobster caught and crosses mark pots that caught no lobster. Commonwealth marine park zones and 10 m depth contours are shown.

Size structure

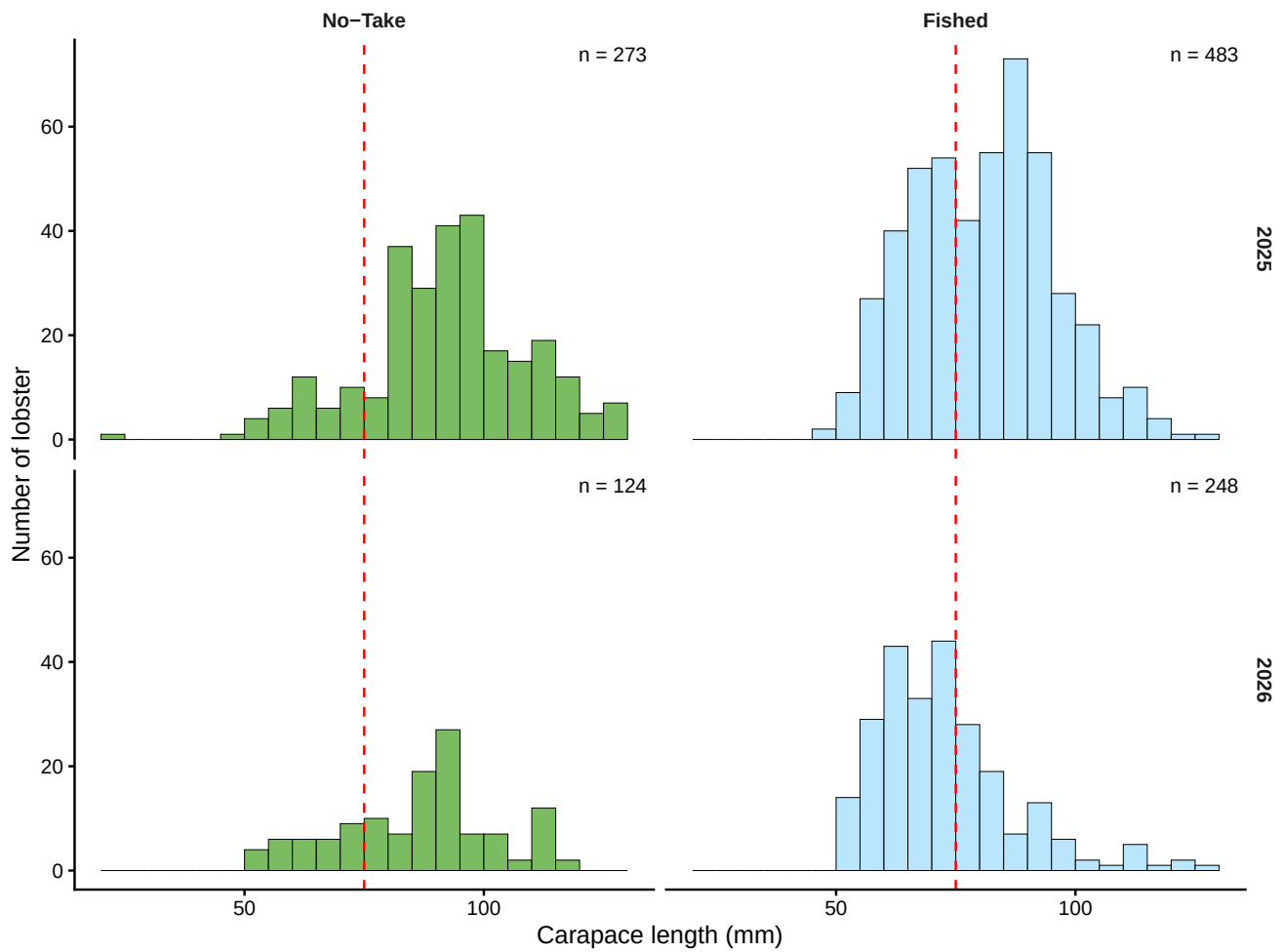


Figure 2: Carapace length frequency distribution of western rock lobster by zone status (columns) and survey year (rows). Bars are counts of lobster in 5 mm length bins. The dashed red line shows the 75 mm carapace length size of maturity.

Abundance by status and year

All lobster

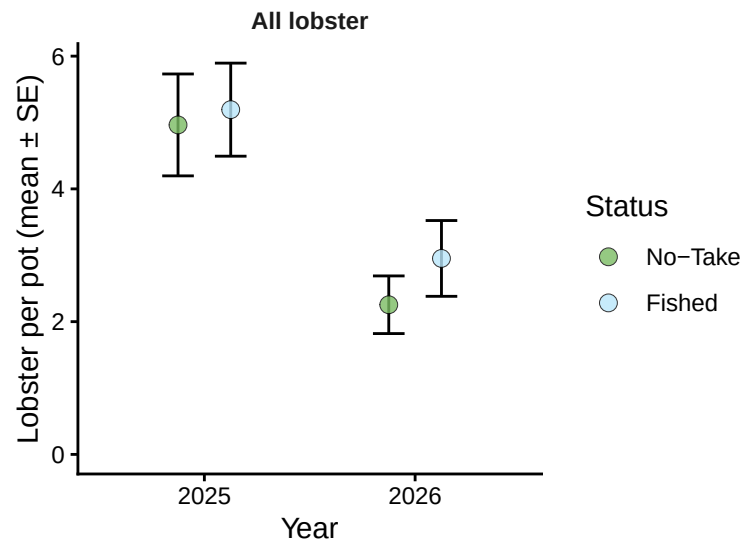


Figure 3: All western rock lobster: mean number of lobster per pot ($\pm$ SE) by zone status and survey year. Pots that caught no lobster are included as zeros.

Sub-size of maturity

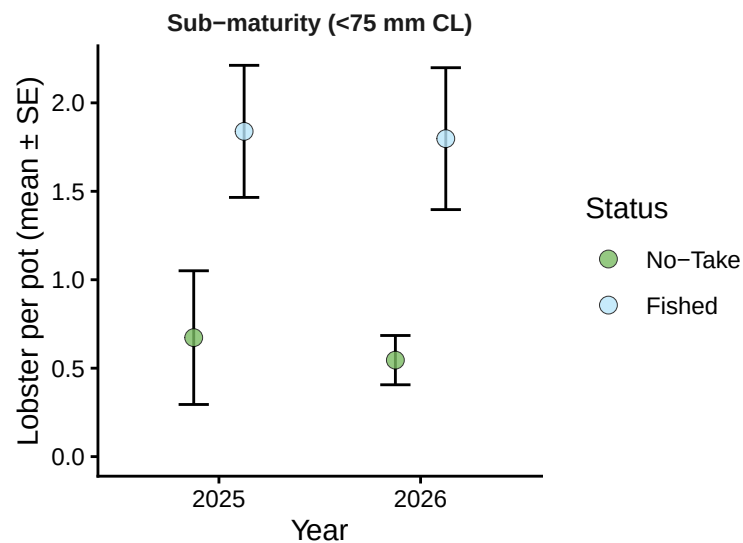


Figure 4: Sub-size of maturity western rock lobster (<75 mm carapace length): mean number of lobster per pot ($\pm$ SE) by zone status and survey year. Pots that caught no lobster are included as zeros.

Greater than size of maturity

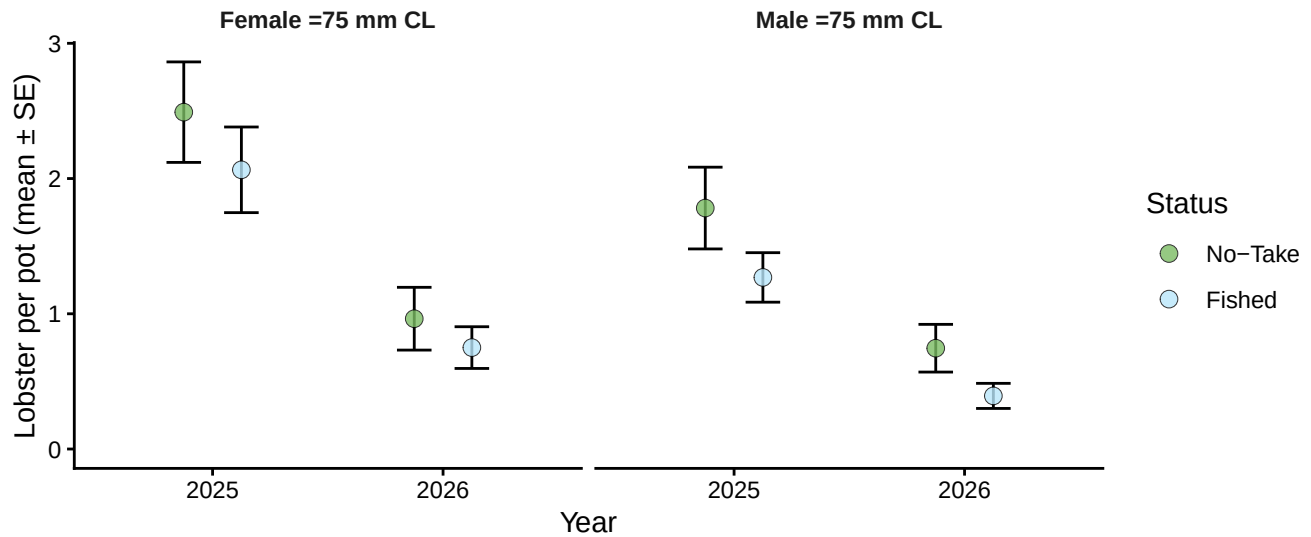


Figure 5: Western rock lobster greater than size of maturity (75 mm carapace length) by sex: mean number of lobster per pot ($\pm$ SE) by zone status and survey year. Lobster without a recorded sex are excluded. Pots that caught no lobster are included as zeros.

30% greater than size of maturity

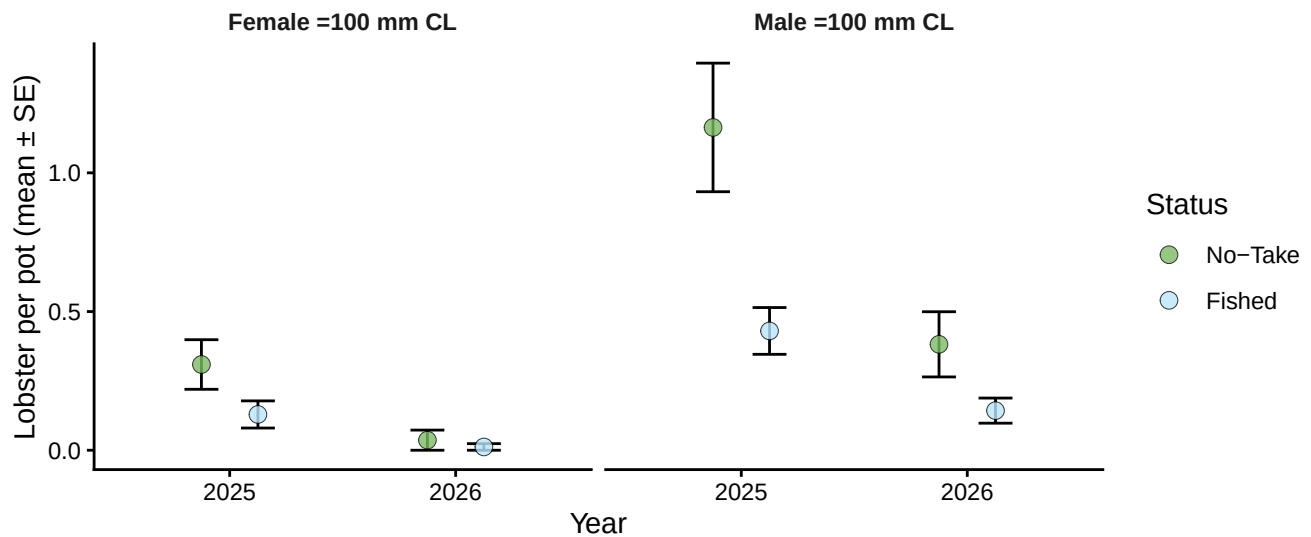


Figure 6: Western rock lobster 30% greater than size of maturity (100 mm carapace length) by sex: mean number of lobster per pot ($\pm$ SE) by zone status and survey year. Lobster without a recorded sex are excluded. Pots that caught no lobster are included as zeros.